

SAT-Style Practice Test

Complete Original Edition

Full-Length Practice Examination

Section 1: Reading and Writing — Modules 1 & 2 (54 Questions)

Section 2: Mathematics — Modules 1 & 2 (44 Questions)

Total: 98 Questions

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SECTION 1, MODULE 1: Reading and Writing

27 Questions — 32 Minutes

Question 1

Skill: Vocabulary in Context

The following text is from Toni Morrison's 1977 novel *Song of Solomon*. The young narrator is describing his daily routine at his family's kitchen table in a midwestern city.

I watched my father from across the breakfast table as he spread marmalade on his toast with slow, deliberate strokes. He rarely acknowledged my presence in the mornings. He was consumed by the newspaper, scanning each headline as if the world's fate depended on it. I did not trouble myself with his silence; it had become as familiar to me as the ticking of the hallway clock.

As used in the text, what does the phrase "trouble myself with" most nearly mean?

- A) Complain about
- B) Investigate
- C) Worry about
- D) Profit from

Question 2

Skill: Words in Context

The fraction of global cropland that a 2024 United Nations report classified as severely degraded surpasses other recent assessments. Given how little _____ there is among agricultural scientists regarding the precise thresholds for degradation categories, this discrepancy should not be surprising: soil researchers frequently challenge one another's classification criteria.

Which choice completes the text with the most logical and precise word or phrase?

- A) funding
- B) cooperation
- C) enthusiasm
- D) agreement

Question 3

Skill: Words in Context

Priya Mehta has shown that international trade sanctions do little to change the likelihood that a targeted nation will alter its domestic labor policies, but sanctions are not as _____ as this finding seems to imply. As Mehta herself observes, sanctions can determine which economic reforms foreign governments prioritize in the first place.

Which choice completes the text with the most logical and precise word or phrase?

- A) controversial
- B) transparent
- C) futile
- D) ineffectual

Question 4

Skill: Words in Context

Though it does not ensure a film's commercial triumph, _____ can play a significant role in that triumph — a well-orchestrated press tour helped make Ridley Scott's *Gladiator* the third-highest-grossing film of 2000.

Which choice completes the text with the most logical and precise word or phrase?

- A) cinematography
- B) promotion
- C) casting
- D) improvisation

Question 5

Skill: Words in Context

Critics sometimes reject an argument simply because the person making it has a financial stake in the outcome, but from a purely logical standpoint, the identity of the arguer is _____: it has no direct bearing on whether the argument is valid.

Which choice completes the text with the most logical and precise word or phrase?

- A) inconclusive
- B) unquestionable
- C) peripheral
- D) irrelevant

Question 6

Skill: Text Structure and Purpose

When most people imagine ancient marine reptiles, they picture streamlined swimmers such as the dolphin-shaped ichthyosaur. However, many marine reptiles that lacked flippers also possessed remarkable aquatic adaptations. For example, recent fossil evidence suggests that the heavy-bodied, short-limbed *Placodus* likely fed on shellfish in shallow coastal waters.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It reveals that the marine reptiles mentioned earlier used their adaptations for feeding.
- B) It gives an example of a marine reptile with the characteristics mentioned earlier in the text.
- C) It defines a term related to marine reptile locomotion mentioned earlier in the text.
- D) It indicates that the marine reptiles mentioned earlier in the text were not the only ones that lived in water.

Question 7

Skill: Text Structure and Purpose

The following text is from James Baldwin's 1953 essay collection *Notes of a Native Son*. The narrator is a young man living in Harlem, New York.

I read voraciously. My mother had enrolled me in the public library as soon as I could print my name on the

application card. I read and catalogued every shelf. I constructed a private geography of stories and located myself within it. At night in bed I designed elaborate quests: Read every book by a single author. Trace a character's journey on a real map. Rewrite the ending of a novel I disliked.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It describes the narrator trying to learn how to print his name.
- B) It provides examples of what the narrator thinks about at night.
- C) It gives directions to the narrator's favorite library.
- D) It portrays the narrator's relationship with his mother.

Question 8

Skill: Main Purpose

Lorraine Hansberry's landmark play *A Raisin in the Sun* premiered in 1959 at the Ethel Barrymore Theatre. Producer Philip Rose, who had spent years working in the music industry, first encountered Hansberry's manuscript through a mutual friend and immediately recognized its power. Together with director Lloyd Richards, Rose assembled a cast that included Sidney Poitier, Ruby Dee, and Claudia McNeil. Since its debut, the Ethel Barrymore Theatre has continued to host productions that amplify underrepresented voices, giving playwrights like August Wilson a prominent stage for their work.

Which choice best states the main purpose of the text?

- A) To show how the music industry influenced Rose and Richards
- B) To bring attention to a specific work by August Wilson
- C) To describe how Rose's and Richards's collaboration evolved over time
- D) To discuss the origin and significance of a historic theater production

Question 9

Skill: Reading Comprehension

The textile industry has long relied on chemical dyes to color fabrics in large batches. Recent breakthroughs in bio-dyeing were achieved through research in South Korea on pigments derived from algae. Bio-dyeing is generally regarded as an improvement over conventional chemical dyeing methods: whereas chemical methods require large volumes of heated water and produce toxic runoff, bio-dyeing uses naturally occurring pigments that bind to fibers at room temperature, thus reducing both energy consumption and wastewater pollution.

Based on the text, what is one disadvantage of some conventional chemical dyeing methods?

- A) They are especially susceptible to supply chain disruptions.
- B) They are only effective on fabrics made from natural fibers.
- C) They generate harmful waste byproducts.
- D) They require more pigment per batch than bio-dyeing does.

Question 10

Skill: Reading Comprehension

"Algebra" is a well-known example of a loanword — that is, a term that originated in one language and was later adopted by another. The word entered English from Medieval Latin *algebra*, which had borrowed it from Arabic *al-jabr*, meaning "reunion of broken parts." Arabic scholars had themselves adapted the mathematical concept

from earlier Greek and Indian traditions. "Algorithm" follows a similar path: it derives from the Latinized name of the Persian mathematician al-Khwarizmi, whose ninth-century treatises were translated into Latin and widely circulated in medieval Europe.

The author makes which point about the Arabic language?

- A) It has served as a conduit through which concepts from other traditions have influenced English.
- B) Its contribution to English vocabulary roughly equals the collective contribution of Latin and Greek.
- C) It has borrowed words from European languages and contributed words to them.
- D) It adopted Greek and Indian words in approximately equal numbers.

Question 11

Skill: Data Interpretation

[Table: Total Enrollment at Five Universities in the United States]

University	Location	Enrollment
State University of New York	New York	81,000
Miami Dade College	Florida	58,000
University of Central Florida	Florida	71,000
Lone Star College	Texas	45,000
Arizona State University	Arizona	77,000

In terms of total enrollment, the University of Texas at Austin is the largest public university in the United States, with roughly 52,000 students. Most public universities are considerably smaller. For example, the total enrollment of Lone Star College in Texas is 45,000 students. And in Florida, the total enrollment of Miami Dade College is _____

Which choice most effectively uses data from the table to complete the statement?

- A) 77,000 students.
- B) 45,000 students.
- C) 81,000 students.
- D) 58,000 students.

Question 12

Skill: Data Interpretation

[Table: Millions of Barrels of Oil Produced in 1990 and 2020]

Country	1990	2020
Norway	1.68	1.78
Brazil	0.65	2.94
Nigeria	1.81	1.35
Colombia	0.44	0.78

While doing research for a paper about global oil production, a student finds a table with information about four different countries. The student notes that the country that produced 2.94 million barrels of oil in 2020 had produced _____

Which choice most effectively uses data from the table to complete the statement?

- A) 0.44 million barrels of oil in 1990.
- B) 1.78 million barrels of oil in 1990.
- C) 1.81 million barrels of oil in 1990.
- D) 0.65 million barrels of oil in 1990.

Question 13

Skill: Command of Evidence (Textual)

"The Yellow Wallpaper" is an 1892 short story written by Charlotte Perkins Gilman. In the story, Gilman suggests that the narrator, a young wife, experiences a transformation in her perception of reality as a result of being confined to a room, observing patterns, writing secretly, _____

Which quotation from "The Yellow Wallpaper" most effectively illustrates the claim?

- A) "John is a physician, and perhaps that is one reason I do not get well faster."
- B) "I sometimes fancy that in my condition if I had less opposition and more society and stimulus."
- C) "She planned a visit to cousin Henry and Julia, delighting in the prospect of company."
- D) "I've got out at last, said I, in spite of you and Jane! And I've pulled off most of the paper, so you can't put me back!"

Question 14

Skill: Data Interpretation

[Table: Average Annual Rainfall in Millimeters, 1960 and 2020]

Region	1960	2020	Change in mm	Percent change
Northern Australia	1,420	1,560	+140	+10%
Southern India	980	870	-110	-11%
Western Sahel	610	530	-80	-13%
Eastern Brazil	1,850	1,610	-240	-13%

A student in an environmental science course is examining shifts in annual rainfall across various regions due to climate variability. The first task is to determine how the decline in Eastern Brazil compares to that in other regions. The student finds that _____

Which choice most effectively uses data from the table to complete the student's conclusion?

- A) the decline in millimeters of rainfall was greater in Eastern Brazil than in Southern India or the Western Sahel.
- B) while rainfall rose in Eastern Brazil between 1960 and 2020, it declined in Southern India and the Western Sahel.
- C) though the percent decrease in rainfall in Southern India was greater than that in Northern Australia, it was less than that in the Western Sahel.
- D) though the percent decrease in rainfall in Eastern Brazil was equal to that in the Western Sahel, the absolute decline was much larger in Eastern Brazil.

Question 15

Skill: Logical Completion

The *Rafflesia arnoldii* flower occurs naturally only in the rainforests of Sumatra and Borneo, and its extremely limited range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to protect individual species most at risk, like *R. arnoldii*, but recently, conservationists have argued that an ecosystem-based approach that accounts for the many interactions among soil fungi, pollinating insects, and host vines of a given forest area may lead to better outcomes for all species in a given location. If this view is correct, the single-species approach to the conservation of *R. arnoldii* could thus _____

Which choice most logically completes the text?

- A) lead to a better understanding of how the distribution of Rafflesia species throughout Southeast Asia has affected their survival.
- B) allow conservationists to better consider how deforestation in Sumatra may reduce the number of species competing with *R. arnoldii*.
- C) mistakenly shift the focus of conservation efforts away from *R. arnoldii* itself.
- D) fail to consider the ways in which the survival of *R. arnoldii* may be influenced by changes in the populations of other organisms that inhabit its habitat.

Question 16

Skill: Standard English Conventions — Verb Form

It is widely recognized that the Smithsonian Institution in Washington, D.C. _____ an extraordinary collection of historical artifacts, including the original Star-Spangled Banner and Charles Lindbergh's *Spirit of St. Louis* aircraft.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) contains
- B) to have contained
- C) containing
- D) to contain

Question 17

Skill: Standard English Conventions — Punctuation

A seasoned mountaineer might summit dozens of peaks over the course of a career, from Denali in Alaska to Kilimanjaro in Tanzania, but no climber has ever achieved the ultimate _____ an unassisted solo ascent of all fourteen eight-thousanders in a single calendar year.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) feat catching
- B) feat. Catching
- C) feat: catching
- D) feat; catching

Question 18

Skill: Standard English Conventions — Punctuation

Today, the Nobel Prize is universally regarded as the pinnacle of scholarly achievement, with its prestigious awards being conferred upon leading researchers like Marie Curie in _____ when Alfred Nobel drafted his will

in 1895, the prizes were largely an afterthought — a way to dispose of his fortune by rewarding those who had conferred the greatest benefit on humankind.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Stockholm; however,
- B) Stockholm,
- C) Stockholm, however,
- D) Stockholm

Question 19

Skill: Standard English Conventions — Punctuation

When attempting to reconstruct an ancient climate record, paleoclimatologists like Dr. Amara Okafor at Columbia University rely in part on data about the width and chemical composition of tree _____ the bristlecone pine forests of the White Mountains in California, the U.S. Forest Service compiles such data in several ways: by partnering with university labs, by linking core samples to dendrochronological databases, and by maintaining a catalog of ring-width measurements spanning over 5,000 years.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) rings for
- B) rings, for
- C) rings. For
- D) rings and for

Question 20

Skill: Standard English Conventions — Verb Tense

In 1953, biochemist Rosalind Franklin produced X-ray diffraction images of DNA that proved critical to understanding its double-helix structure. Years later, she _____ for this pioneering work with widespread recognition from the scientific community, and her contributions would eventually be cited in countless genetics textbooks.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) had been recognized
- B) would be recognized
- C) is recognized
- D) will be recognized

Question 21

Skill: Standard English Conventions — Punctuation

For Japan, a member of the G7 group of industrialized nations since 1975, the alliance's framework of economic coordination confers both benefits and _____ organization's many members, nations as disparate as Germany and Italy, are all committed to supporting Japan's trade interests, the reverse is also expected.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) responsibilities: while the
- B) responsibilities; the

- C) responsibilities, while the
- D) responsibilities. The

Question 22

Skill: Transitions

If you could dive to the floor of the Mariana Trench, the deepest point in the ocean, you would find organisms adapted to extreme pressure. This means that life in the trench thrives under conditions that would crush most surface-dwelling species. _____ at hydrothermal vents along the Mid-Atlantic Ridge, you would find a contrasting example of deep-sea adaptation, with organisms relying on chemosynthesis rather than pressure resistance to survive.

Which choice completes the text with the most logical transition?

- A) For example,
- B) Specifically,
- C) In other words,
- D) Elsewhere,

Question 23

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Archaeologists have long debated the purpose of the Nazca Lines in Peru's coastal desert.
- A 2015 study proposed that the geoglyphs served as ceremonial pathways for water-related rituals.
- In 2019, a different study used satellite imaging to show that many lines align precisely with underground aqueducts.
- That study concluded that the lines functioned as markers guiding travelers to water sources.

The student wants to describe how scientific understanding about the lines' purpose has evolved. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Satellite imaging revealed that the Nazca Lines align with underground aqueducts, according to recent analysis.
- B) A 2015 study proposed that the Nazca Lines served a ceremonial purpose, but newer research suggests they functioned as practical guides to water sources.
- C) Archaeologists have long studied the Nazca Lines, with one study concluding they were ceremonial pathways through the desert.
- D) Studies in 2015 and 2019 offered different explanations for the purpose of the Nazca Lines in Peru.

Question 24

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Alexander Calder is an American artist known for his sculptures and large-scale public artworks.
- He gained fame in 1931 with his first kinetic sculptures, known as mobiles.
- He completed *Flamingo* in 1974.
- It is a stabile — a stationary sculpture — made of steel, standing 53 feet tall in a Chicago plaza.
- He completed *Flying Colors* in 1975.
- It is a mobile — a hanging kinetic sculpture — made of aluminum and steel wire, displayed indoors.

Which choice most effectively uses information from the given sentences to emphasize a difference between *Flamingo* and *Flying Colors*?

- A) Calder's *Flamingo* is a stationary steel stabile; *Flying Colors*, by contrast, is a kinetic mobile made of aluminum and wire.
- B) Alexander Calder is known for his sculptures, such as *Flamingo* and *Flying Colors*.
- C) After gaining fame with his mobiles, Alexander Calder completed *Flamingo*, another large-scale work.
- D) The monumental scale of Calder's mobiles is reflected in *Flying Colors*, a hanging sculpture of aluminum and steel wire.

Question 25

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The Chihuahuan Desert grasslands are a biome found in the southwestern United States and northern Mexico.
- The grasslands are one of several "biodiversity hotspots" identified by Conservation International.
- A biodiversity hotspot is a region with an exceptionally high number of endemic species that is also experiencing significant habitat loss.
- Conservation International stated, "These regions contain at least 1,500 species of vascular plants found nowhere else."
- The organization also noted, "Each hotspot has lost at least 70% of its original natural vegetation."

The student wants to explain what a biodiversity hotspot is. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Chihuahuan Desert grasslands are a biodiversity hotspot in the southwestern United States whose native species are found nowhere else on Earth.
- B) A biodiversity hotspot is a region, such as the Chihuahuan Desert grasslands, that has an exceptionally high number of endemic species and is experiencing significant habitat loss.
- C) The Chihuahuan Desert grasslands, which Conservation International considers a biodiversity hotspot, are located in the southwestern United States.
- D) Conservation International considers the Chihuahuan Desert grasslands to contain a "unique ecosystem."

Question 26

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The Olympic Games are a series of international athletic competitions held every four years.
- A gold medal indicates that an athlete finished in first place.
- Simone Biles competed as an Olympic gymnast from 2016 to 2024.
- She won eleven medals, seven of which were gold.
- Michael Phelps competed as an Olympic swimmer from 2000 to 2016.
- He won twenty-eight medals, twenty-three of which were gold.

Which choice most effectively uses information from the given sentences to emphasize the number of gold medals won by Michael Phelps?

- A) In his career as an Olympic swimmer, Michael Phelps won twenty-eight total medals — twenty-three of them were gold.

B) Both multiple gold medalists, Simone Biles and Michael Phelps debuted at the Olympics in 2016 and 2000, respectively.

C) At the Olympic Games, a gold medal indicates a first-place finish, and some accomplished athletes like Simone Biles have won many.

D) An accomplished Olympian with twenty-eight total medals, Michael Phelps competed as a swimmer from 2000 to 2016.

Question 27

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The human nose contains olfactory receptors that detect airborne chemical compounds.
- The sensation of spiciness from chili peppers is caused by capsaicin binding to pain receptors, not olfactory receptors.
- Participants in a study inhaled the aroma of freshly ground black pepper.
- They rated its pungency as strong.
- The participants inhaled the aroma of freshly ground white pepper.
- They rated its pungency as mild.

The student wants to make a generalization about ground pepper. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Black pepper and white pepper both contain capsaicin, which binds to pain receptors.

B) In a research study, participants inhaled the aromas of black pepper and white pepper, two common spices.

C) Participants in a study rated the pungency of various spices.

D) Different varieties of ground pepper can vary in perceived pungency.

SECTION 1, MODULE 2: Reading and Writing

27 Questions — 32 Minutes

Question 1

Skill: Words in Context

Although Zora Neale Hurston's 1937 novel *Their Eyes Were Watching God* and Nella Larsen's 1929 novel *Passing* may have had some _____ at the time of their initial publication, their standing in literary scholars' estimations is now firmly established. In 2020, for instance, critics for *The Guardian* described the former as "luminous" and the latter as "masterfully restrained."

Which choice completes the text with the most logical and precise word or phrase?

- A) advocates
- B) skeptics
- C) forerunners
- D) detractors

Question 2

Skill: Words in Context

Behavioral economists have found that although consumers claim to prefer brands that offer transparent pricing and demonstrate social responsibility, when asked to compare advertisements emphasizing such values with advertisements expressing _____ claims, consumers tend to respond to the bold certainty of the latter more favorably.

Which choice completes the text with the most logical and precise word or phrase?

- A) assertive
- B) indecisive
- C) modest
- D) deceptive

Question 3

Skill: Words in Context

A number of contemporary chefs _____ boundaries between traditional regional cuisines and modern fusion cooking. For example, Korean-American chef David Chang incorporated elements of his grandmother's recipes into avant-garde tasting menus at his restaurant Momofuku, and Peruvian chef Virgilio Martinez combined Andean ingredients with molecular gastronomy techniques at his Lima restaurant Central.

Which choice completes the text with the most logical and precise word or phrase?

- A) reinforce
- B) blur
- C) acknowledge
- D) fabricate

Question 4

Skill: Main Purpose

Today, architect Frank Lloyd Wright is primarily celebrated for his iconic residential designs like Fallingwater and the Robie House. However, by overlooking his lesser-known projects, admirers will miss the full scope of Wright's innovation. For instance, his Johnson Wax Headquarters and the V.C. Morris Gift Shop skillfully merge commercial functionality with organic architecture. These buildings deserve as much recognition as Wright's most famous houses.

Which choice best states the main purpose of the text?

- A) To argue that more attention should be given to Wright's lesser-known works
- B) To describe the similarities and differences between residential and commercial architecture
- C) To explain how Wright learned to design commercial buildings
- D) To encourage architects to study many different building types

Question 5

Skill: Main Purpose

Moby-Dick, first published in 1851, is a novel in Herman Melville's body of work, which includes more than a dozen completed books. Some critics have found fault with the lengthy cetological digressions in *Moby-Dick* and other novels in the series of sea narratives, saying that they interrupt the plot and test the reader's patience. But other critics argue that the books should not be judged as conventional novels but as a single ambitious literary experiment, similar to other multivolume projects such as Marcel Proust's *In Search of Lost Time*.

Which choice best states the main purpose of the text?

- A) To explain why many critics find Melville's novels entertaining despite structural flaws
- B) To describe a characteristic of Melville's novels and offer two differing viewpoints on this characteristic
- C) To argue that Melville's sea narratives should have the literary renown of Proust's work
- D) To present a reason the unusual structure Melville uses for *Moby-Dick* makes it his most complex novel

Question 6

Skill: Cross-Text Connections

Text 1

Emily Dickinson, author of *Fascicle 40* and numerous uncollected poems, is highly regarded despite her unconventional style. Her syntax is fragmented; her poems rely on dashes rather than standard punctuation. That her work is compelling despite this is a testament to her extraordinary insight — even if readers engage with her poems only for the ideas, they will find no shortage of profound observations.

Text 2

Dickinson is sometimes criticized for her style, but it is wrong to fault a poet for refusing to do what she never intended to do. For example, although most of her poems consist of compressed imagery and abrupt line breaks and we learn little about narrative continuity, Dickinson wanted to capture fleeting moments of perception, and one of her points is that at such a scale, conventional grammar does not apply. Thus her unconventional punctuation is central to her thematic aims.

Based on the texts, how would the author of Text 2 most likely respond to the evaluation of Dickinson's writing style in the underlined claim in Text 1?

- A) By asserting that Dickinson's unconventional punctuation is likely a deliberate artistic choice that conveys an important idea
- B) By arguing that Dickinson's style should be judged against the conventions of essayists rather than those of other poets
- C) By agreeing that Dickinson's syntax is fragmented but disagreeing about the quality of her imagery
- D) By pointing out that the dash-heavy style is a feature of *Fascicle 40* but not of her uncollected poems

Question 7

Skill: Command of Evidence (Inference)

Tomoko Sato and colleagues relied on ancient DNA (aDNA) — genomic material incidentally preserved in archaeological bone fragments housed in museum collections — to investigate the migratory patterns of the prehistoric Jomon people of Japan. Although this approach offers unique advantages, such as the ability to trace population movements across millennia, it remains a relatively underutilized resource because aDNA is often to some extent degraded, a situation not easily remedied with current extraction and sequencing technologies.

Information in the text best supports which statement about aDNA?

- A) It may yield insights that other types of genomic data cannot.
- B) It may be underused because of its controversial status among scientists.
- C) It has thus far proved valuable mainly to researchers studying ancient plants.
- D) It tends to be much more degraded than other types of DNA of comparable age.

Question 8

Skill: Main Idea

Rachel Carson and colleagues have explored how biomagnification — a process that occurs when a toxin accumulates at increasing concentrations up a food chain — can result from pesticide exposure shared across trophic levels. Meanwhile, Theo Colborn and colleagues have investigated how biomagnification operates through different biochemical pathways, but the relative prevalence of biomagnification through shared and different pathways is still poorly understood. This motivated ecologists David Schindler and Jules Blais to evaluate both types of biomagnification in a single study for their 2007 paper.

Which choice best states the main idea of the text?

- A) Schindler and Blais's study was conducted using data from the studies by Carson and Colborn.
- B) Schindler and Blais's study addresses biomagnification more comprehensively than the studies by Carson and Colborn do.
- C) Both the study by Carson and that by Schindler and Blais compare biomagnification through shared pathways to biomagnification through different pathways.
- D) Before Schindler and Blais's study, biomagnification was assumed to require a shared biochemical mechanism between two trophic levels.

Question 9

Skill: Command of Evidence (Strengthen/Weaken)

Over time, stalactites in limestone caves grow incrementally as mineral-rich water drips from the ceiling, and the chemical composition of each layer is influenced by the prevailing climate conditions when the layer forms. Geochemist Lucia Ferraro and colleagues analyzed the isotopic ratios of oxygen-18 in cores from two

stalactites — one from a chamber at 50 meters depth (sample A) and one from a chamber at 200 meters depth (sample B) — finding that sample A's oxygen-18 concentration matched that of rainwater from approximately 800 years ago and that sample B's concentration matched rainwater from approximately 2,400 years ago. Ferraro et al. concluded that sample A formed during a more recent climatic period than did sample B.

Which finding, if true, would most directly support the researchers' conclusion?

- A) In stalactites, growth rate predicts layer thickness more reliably than does oxygen-18 concentration.
- B) Regardless of depth, stalactites in the same cave system show identical oxygen-18 concentrations.
- C) Water chemistry at 50 meters depth tends to reflect more recent precipitation than does water chemistry at 200 meters depth.
- D) Among stalactites, there is no correlation between depth and the age of the mineral deposits.

Question 10

Skill: Command of Evidence (Strengthen/Weaken)

Elena Voss, Henrik Johansson, and colleagues assembled a database of carbon emissions from manufacturing plants by auditing facilities in the field and by gathering data from similar published audits. The team gave each of these audits a rating for transparency (with a higher rating for clearer descriptions of measurement methods) and for standardization (with a higher rating for using units more closely aligned with international reporting protocols). Based on these ratings, a second research team concluded that the measurement methodology was better documented in an audit by Kimura and Tanaka than it was in an audit by Petrov and colleagues.

Which finding, if true, would most directly challenge the second research team's conclusion?

- A) The audit by Kimura and Tanaka had a lower standardization rating than the audit by Petrov and colleagues did.
- B) The audit by Kimura and Tanaka had a high standardization rating and a high transparency rating.
- C) The audit by Petrov and colleagues had a medium standardization rating and a medium transparency rating.
- D) The audit by Kimura and Tanaka had a lower transparency rating than the audit by Petrov and colleagues did.

Question 11

Skill: Command of Evidence (Strengthen/Weaken)

Urban green spaces in temperate cities have been shown to reduce ambient temperatures in response to rising heat, potentially enabling increased carbon dioxide absorption through expanded tree canopy cover, but also potentially enabling increased methane emissions from decomposing organic matter in waterlogged park soils. Urban ecologist Mei-Ling Chen and her colleagues developed a model incorporating numerous variables — soil moisture levels, canopy density data, and seasonal temperature fluctuations among them — to evaluate the net effect of green spaces on greenhouse gas levels in cities, concluding that net emissions are likely to increase if warmer temperatures accelerate soil decomposition.

Which finding, if true, would most directly support the researchers' conclusion?

- A) Warmer temperatures reduce soil moisture in urban parks, limiting both tree growth and methane-producing decomposition.
- B) Warmer temperatures extend the growing season for urban trees but have no effect on the rate of soil decomposition.
- C) Warmer temperatures slow the growth of urban tree species while also suppressing soil decomposition.

D) Warmer temperatures amplify the effect of soil moisture on both canopy growth and the rate of organic matter decomposition.

Question 12

Skill: Data Interpretation (Graph)

[Graph: "Female Elected Officials in National Legislatures, 2010–2014" — Line graph tracking three countries (Peru, South Korea, and Finland) from 2010 to 2014. Finland's line (circles, dotted) stays steady around 85 officials across all years. Peru's line (squares, dashed) rises from about 28 in 2010 to about 40 in 2014. South Korea's line (triangles, solid) stays low, hovering between 15 and 20 throughout.]

A report from an international organization that monitors the number of female elected officials shows that between 2010 and 2014 _____

Which choice most effectively uses data from the graph to complete the example?

- A) Peru was lower in 2014 than in 2010, whereas the number in Finland was the same in 2014 as in 2010.
- B) South Korea was greater in 2014 than in 2010, whereas the number in Finland was the same in 2014 as in 2010.
- C) South Korea was greater in 2014 than in 2010, whereas the number in Peru was lower in 2014 than in 2010.
- D) Finland was the same in 2014 as in 2010, but it had more female officials than either South Korea or Peru did in 2014.

Question 13

Skill: Logical Completion

The German Socio-Economic Panel (GSOEP) is a longitudinal study surveying approximately 14,000 households in Germany to gather extended data on income inequality trends. To carry out the necessary repeated sampling of the same population over many years, GSOEP requires substantial institutional funding, but the method provides unique insights into causal relationships. However, when questions of causation are beside the point, as with a traffic study seeking only to count the number of vehicles per hour using a city's highway system, longitudinal methods are unnecessary, and so _____

Which choice most logically completes the text?

- A) the expense of GSOEP is likely greater than the cost of longitudinal traffic studies.
- B) conclusions drawn from GSOEP are likely to be more authoritative than those from the traffic study.
- C) longitudinal methods are suitable for income studies but ought to be avoided for traffic studies.
- D) the success of the traffic study likely requires significantly less financial outlay than that needed for GSOEP.

Question 14

Skill: Logical Completion

The many recordings of Ludwig van Beethoven's 1808 *Symphony No. 5* are not sonically identical, and musicologists debate which performances best capture Beethoven's intended tempos. One no longer widely heard interpretation is a 1962 recording by conductor Herbert von Karajan, who followed mid-twentieth-century performance traditions favoring slower tempos and heavier orchestration rather than the historically informed practices gaining favor at the time. This recording was later found to have significant deviations from

Beethoven's metronome markings. However, few musicologists of that era had expertise in both historical performance practice and orchestral interpretation, and most of those who did had worked with von Karajan. So, it is unsurprising that initial critical reviews of the 1962 recording were mostly _____

Which choice most logically completes the text?

- A) negative, since any musicologist with expertise in both historical and modern practice most likely collaborated with von Karajan and would therefore not review it.
- B) negative, since those musicologists with the necessary expertise would be aware that metronome markings cannot reliably indicate intended tempo.
- C) positive, since scholars who reviewed the recording were unaffiliated with von Karajan and were mostly specialists who lacked expertise in historical performance practices.
- D) positive, since the recording followed conventions that most mid-century critics were comfortable with.

Question 15

Skill: Logical Completion

Stories about King Arthur derive from many sources, such as Geoffrey of Monmouth's *Historia Regum Britanniae*, composed in the 1130s, and the Welsh *Mabinogion* from the 13th century. One of the most significant later sources, the French poet Chrétien de Troyes's *Lancelot*, was written around 1177; some of its material was adapted by the German poet Wolfram von Eschenbach into *Parzival* around 1210. But while no source before 1177 includes references to the quest for the Holy Grail, both Chrétien's *Lancelot* and Sir Thomas Malory's 15th-century *Le Morte d'Arthur* do. It can therefore be inferred that _____

Which choice most logically completes the text?

- A) *Le Morte d'Arthur* is more historically accurate than *Historia*, because the *Mabinogion* had not been written when Geoffrey was composing his work.
- B) Geoffrey of Monmouth's accounts of Arthurian legends in his *Historia* are more similar in content to the accounts in the *Mabinogion* than to those in *Lancelot*.
- C) Malory did not use the *Mabinogion* as a source for information he presented about the Grail quest.
- D) Geoffrey of Monmouth was unaware of the Grail legend when composing his *Historia*, though works containing such stories were available to later authors.

Question 16

Skill: Standard English Conventions — Punctuation

In the Philippines, it is not uncommon for a province to be known by a nickname that corresponds to one of its notable features, like topography, culture, or principal crops. For example, the province of Cebu has also been called "the Queen City of the South," a title that alludes to what the region is well _____ the vibrant festivals and trade that have defined its identity for centuries.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) known: for
- B) known for:
- C) known for
- D) known for,

Question 17

Skill: Standard English Conventions — Verb Form

A territory of the Ottoman Empire encompassing the present-day city of Thessaloniki, Greece, the multilingual province of Salonica was in fact only nominally autonomous. The Young Turk revolution of 1908, by empowering reformists and other factions within the empire to assert their political ambitions, _____ to the empire's eventual dissolution.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) which contributed
- B) contributed
- C) having contributed
- D) contributing

Question 18

Skill: Standard English Conventions — Punctuation

Gemstones can be classified by how they interact with light. For example, nearly all light passes through diamond, which is classified as _____ only some light to pass through, jade is instead classified as translucent.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) transparent, allowing
- B) transparent allowing
- C) transparent. Allowing
- D) transparent and allowing

Question 19

Skill: Standard English Conventions — Verb Form

The sculptor Auguste Rodin's masterpiece *The Thinker*, symbolically _____ to represent the power of intellectual contemplation that defines the human spirit, is a widely celebrated example of Romantic-era sculpture.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) functioning
- B) has functioned
- C) functions
- D) was functioning

Question 20

Skill: Standard English Conventions — Punctuation

The interaction between chromosomes and epigenetic markers reveals how cells with identical genetic material develop different _____ whereas the chromosome in each cell contains a complete set of genes, epigenetic markers consist of chemical tags that determine which genes in the sequence will be activated.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) characteristics:
- B) characteristics and,
- C) characteristics

D) characteristics,

Question 21

Skill: Standard English Conventions — Punctuation

That the highest point on Earth's surface is located in the Himalayan mountain range of Nepal was accepted by all _____ pinpointing its precise elevation, however, proved more contentious.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) geographers,
- B) geographers;
- C) geographers:
- D) geographers

Question 22

Skill: Transitions

The Colossal Head sculptures of the ancient Olmec civilization are among the most recognizable artifacts of Mesoamerican art. One such head, known as San Lorenzo Monument 1, was discovered at the San Lorenzo archaeological site in Veracruz, Mexico. _____ at the La Venta site in the neighboring state of Tabasco, another colossal head designated as Monument 4 was unearthed.

Which choice completes the text with the most logical transition?

- A) As a result,
- B) Specifically,
- C) For example,
- D) Elsewhere,

Question 23

Skill: Transitions

In the Southeast Asian nation of Vietnam, the percentage of the population living in urban areas increased from 30.4% to 37.3% between 2010 and 2020; this upward trend in urbanization was observed across the region as a whole. _____ from 2010 to 2020, the overall percentage of the Southeast Asian population living in cities rose from 44.1% to 49.9%.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) Indeed,
- C) In reality,
- D) Later,

Question 24

Skill: Transitions

Brazilian musicologist Hermano Vianna argued that samba music is best understood not as a monolithic genre but as a mosaic of diverse regional traditions (because samba incorporated the contributions of a wide range of Afro-Brazilian, European, and Indigenous musical forms). _____ Vianna titled his influential academic study of

samba's origins and cultural significance *The Mystery of Samba*.

Which choice completes the text with the most logical transition?

- A) In sum,
- B) To that end,
- C) In addition,
- D) That is,

Question 25

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- A patent protects an inventor's rights to an invention for a limited period.
- When a patent expires, the invention enters the public domain.
- Public domain inventions can be freely manufactured by anyone.
- The Wright Brothers' flying machine was patented in 1906.
- The patent expired in 1923 (seventeen years later).

Which choice most effectively uses information from the given sentences to explain why the Wright Brothers' flying machine patent entered the public domain?

- A) Once the flying machine patent entered the public domain, anyone could manufacture the design.
- B) The Wright Brothers' flying machine, patented in 1906, is freely available to the public.
- C) Before the flying machine patent entered the public domain, only the Wright Brothers could authorize its production.
- D) The patent on the Wright Brothers' flying machine expired in 1923, causing the invention to enter the public domain.

Question 26

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Tlacolula de Matamoros is a municipality in the state of Oaxaca, Mexico.
- Municipalities are governmental units responsible for delivering many public services to their residents.
- One service they provide is road maintenance.
- Tlacolula's population was 21,082 in 2020.
- Oaxaca is divided into 570 municipalities.

The student wants to provide a specific example of what municipalities provide. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Tlacolula de Matamoros is one of 570 municipalities in Oaxaca, Mexico.
- B) In 2020, the municipality of Tlacolula had a population of 21,082.
- C) The municipality Tlacolula de Matamoros in Oaxaca is responsible for providing many public services.
- D) Road maintenance is just one of the many public services that municipalities provide.

Question 27

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The Mohs scale of mineral hardness is a ten-point scale that ranks minerals by their ability to scratch other minerals.
- Minerals with higher numbers are harder and can scratch those with lower numbers.
- Minerals with lower numbers are softer and cannot scratch those with higher numbers.
- The mineral quartz has a Mohs number of 7.
- The mineral corundum has a Mohs number of 9.
- The mineral feldspar has a Mohs number of 6.

The student wants to compare the hardness of quartz and corundum. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A mineral with a Mohs number of 9, like corundum, is harder than one with a Mohs number of 7, like quartz.
- B) The Mohs scale can be used to order corundum, quartz, and feldspar by their ability to scratch other minerals.
- C) Corundum can leave visible scratches on quartz, which means that corundum is harder than quartz.
- D) Corundum can leave visible scratches on feldspar, which is why corundum has a higher Mohs number.

SECTION 2, MODULE 1: Mathematics

22 Questions — 35 Minutes

Question 1

Skill: Linear Models / Scatterplots

[Scatterplot showing the relationship between x and y . Data points: approximately (1, 5), (2, 7), (2.5, 8), (3, 9), (3.5, 9), (4, 10), (5, 11), (6, 12), (7, 14), (8, 15). A line of best fit passes through approximately (0, 3) and (8, 15).]

Which of the following equations best represents the line of best fit shown?

- A) $y = 3 + 1.5x$
- B) $y = 1.5 + 3x$
- C) $y = 5 + x$
- D) $y = 2 + 2x$

Question 2

Skill: Systems of Linear Equations

$$5x - 3y = 11$$

$$x = 4$$

What is the solution (x, y) to the given system of equations?

- A) (4, 3)
- B) (4, 9)
- C) (4, -3)
- D) (4, -9)

Question 3

Skill: Linear Equations in One Variable

What value of x is the solution to the equation $15x + 36 = 24x$?

- A) $\frac{1}{4}$
- B) $\frac{4}{3}$
- C) 4
- D) 9

Question 4

Skill: Function Evaluation

The function g is defined by $g(x) = \left(\frac{3}{5}\right)x + 52$. What is the value of $g(30)$?

Answer: 70

[Student-Produced Response]

Question 5

Skill: Function Evaluation

The function f is defined by $f(x) = 5 + \sqrt{x}$. What is the value of $f(49)$?

- A) 0
- B) 12
- C) 17
- D) 26

Question 6

Skill: Geometry — Congruent Triangles

Triangles PQR and STU are congruent, where P corresponds to S, and Q and T are right angles. The measure of angle P is 34° . What is the measure, in degrees, of angle U?

Answer: 56

[Student-Produced Response]

Question 7

Skill: Probability

At a music festival, there are a total of 480 attendees. Each attendee is in either Hall X, Hall Y, or Hall Z. If one of these attendees is selected at random, the probability of selecting someone in Hall X is 0.45, and the probability of selecting someone in Hall Y is 0.30. How many attendees are in Hall Z?

- A) 24
- B) 120
- C) 144
- D) 216

Question 8

Skill: Linear Function Interpretation

$$h(t) = -0.04t + 12.8$$

The given function h models the number of liters of water remaining in a tank after draining for t minutes. According to the model, about how many liters of water are drained each minute?

- A) 0.04
- B) 12.8
- C) 32
- D) 320.0

Question 9

Skill: Geometry — Circle Equation

What is the center of the circle in the xy -plane defined by the equation $(x - 7)^2 + (y + 3)^2 = 49$?

- A) $(-7, -3)$
- B) $(-7, 3)$

- C) (7, -3)
- D) (7, 3)

Question 10

Skill: Linear Equations

If $3x + 5 = 14$, what is the value of $6x - 7$?

Answer: 11

[Student-Produced Response]

Question 11

Skill: Exponential Functions — Interpretation

$$p(x) = 8,500(0.82)^x$$

A wildlife biologist implemented a program to manage a certain invasive fish population in a lake. The given function estimates this species' population x years after 2010, where $x \leq 10$. Which of the following is the best interpretation of 8,500 in this context?

- A) The estimated percent decrease in the fish population each decade after 2010
- B) The estimated percent decrease in the fish population each year after 2010
- C) The estimated fish population 10 years after 2010
- D) The estimated initial fish population in 2010

Question 12

Skill: Linear Equations — Slope

Line p is defined by the equation $3x - 4y = 12$. Line q is parallel to line p in the xy -plane. What is the slope of line q ?

- A) $\frac{4}{3}$
- B) $\frac{3}{4}$
- C) -3
- D) -4

Question 13

Skill: Linear Equations — Word Problems

The cost to hire a catering service from Company Z is \$1,200 for the first 3 hours and an additional \$75 per hour for each hour after the first 3 hours. If the total cost to hire the catering service for t hours, where $t > 3$, is \$1,650, which equation represents this situation?

- A) $1200(t - 3) + 75t = 1,650$
- B) $1200(3t) + 75t = 1,650$
- C) $1200 + 75(t - 3) = 1,650$
- D) $1200 + 75(3t) = 1,650$

Question 14

Skill: Quadratic Functions — Interpretation

$$y = -(1/3)x^2 + 2x + 24$$

The given equation models a company's active contracts over 6 months, where y is the estimated number of active contracts x months after the end of March 2015, where $0 \leq x \leq 6$. Which statement is the best interpretation of the y -intercept of the graph?

- A) At the end of March 2015, the estimated number of active contracts was 0.
- B) At the end of March 2015, the estimated number of active contracts was 24.
- C) At the end of April 2015, the estimated number of active contracts was 0.
- D) At the end of April 2015, the estimated number of active contracts was 24.

Question 15

Skill: Trigonometry — Complementary Angles

In right triangle MNP, the sum of the measures of angle M and angle N is 90 degrees. The value of $\sin(M)$ is $(5\sqrt{3})/11$. What is the value of $\cos(N)$?

- A) $(5\sqrt{3})/11$
- B) $(5\sqrt{3})/9$
- C) $(9\sqrt{3})/10$
- D) $(11\sqrt{3})/12$

Question 16

Skill: Linear Equations — Word Problems

$$24.5x + 37.25y = 518$$

Marco ordered gravel and sand, which cost a total of \$518, for his garden. The given equation represents the relationship between the number of bags of gravel, x , and the number of bags of sand, y , Marco ordered. How much more, in dollars, did a bag of sand cost Marco than a bag of gravel?

Answer: 12.75

[Student-Produced Response]

Question 17

Skill: Geometry — Volume

A hemisphere is half of a sphere. If a hemisphere has a radius of 40 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A) 2,700
- B) 10,700
- C) 134,000
- D) 268,100

Question 18

Skill: Geometry — Right Triangles

In triangle DEF, angle E is a right angle and the length of EF is 300 millimeters. If $\cos D = 3/5$, what is the length, in millimeters, of DE?

- A) 75
- B) 180
- C) 225
- D) 375

Question 19

Skill: Exponential Functions — Modeling

In a certain region, the population of a migratory butterfly species is estimated each spring by counting the number observed along designated routes. On average, each year from 2008 to 2018 the count per route decreased by 4.2% of the count the previous year. Based on this average, if there were 8.15 butterflies per route in 2008, which of the following best approximates the count per route in 2018?

- A) $0.958(8.15)^{10}$
- B) $1.042(8.15)^{10}$
- C) $8.15(0.042)^{10}$
- D) $8.15(0.958)^{10}$

Question 20

Skill: Problem Solving — Density

To study structural properties of bamboo, cylindrical samples were cut from 30 stalks. The length of one sample is 3.00 centimeters and its diameter is 2.00 centimeters (radius = 1.00 cm). If this cylinder has a mass of 4.71 grams, what is the density of this sample, in grams per cubic centimeter? (Volume of a cylinder = π times r -squared times h ; use $\pi = 3.14$)

Answer: 0.50

[Student-Produced Response]

Question 21

Skill: Linear Equations — Tables

[Table: $x = -8$, $y = -30$ and $x = 12$, $y = 50$]

The table shows two values of x and their corresponding values of y . The graph of the linear equation representing this relationship passes through the point $(1/2, a)$. What is the value of a ?

Answer: 4

[Student-Produced Response]

Question 22

Skill: Systems of Linear Equations — No Solution

Which system of linear equations has no solution?

- A) $-3x + 6y = -9$ and $3x - 6y = 9$

B) $3x - 6y = 9$ and $5x + 7y = 4$

C) $3x - 6y = 9$ and $-9x + 18y = -27$

D) $-3x + 6y = 9$ and $6x - 12y = 24$

SECTION 2, MODULE 2: Mathematics

22 Questions — 35 Minutes

Question 1

Skill: Functions — Graph Reading

[Graph of $y = f(x)$ in the xy -plane showing an exponential-looking curve. The curve passes through $(-2, 1)$, $(0, 3)$, and $(2, 7)$, with a horizontal asymptote near $y = 0$ for large negative x .]

The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0)$?

- A) 5
- B) 3
- C) 0
- D) -3

Question 2

Skill: Polynomials — Addition

Which expression is equivalent to $(7ab^2 + 4ab) + (3a^2b + 6ab)$?

- A) $10a^2b^2 + 10ab$
- B) $10a^3b^3 + 10a^2b^2$
- C) $42a^3b^3 + 18a^2b^2$
- D) $7ab^2 + 3a^2b + 10ab$

Question 3

Skill: Ratios and Proportions

If $(2c)/d = 7.2$ and $c/(dn) = 28.8$, what is the value of $1/n$?

Answer: 8

[Student-Produced Response]

Question 4

Skill: Linear Equations — Rearranging

$$u = -w/(243z)$$

The given equation relates the distinct positive numbers u , w , and z . Which equation correctly expresses w in terms of u and z ?

- A) $w = -243uz$
- B) $w = -243u/z$
- C) $w = -z/(243u)$
- D) $w = u + 243z$

Question 5

Skill: Percentages — Word Problems

A tutoring center offers a series of three workshops. 800 people attended the first workshop. 55% of those who attended the first workshop attended the second workshop, and 40% of those who attended both the first and second workshops attended the third workshop. How many people attended all three workshops?

Answer: 176

[Student-Produced Response]

Question 6

Skill: Functions — Transformations

[Graph of $y = f(x)$, a straight line passing through $(-4, -5)$ and $(2, 1)$, with a y-intercept at $(0, -1)$.]

The graph of $y = f(x)$ is shown. What is the y-intercept of the graph of $y = f(x) + 8$?

- A) $(0, -9)$
- B) $(0, 7)$
- C) $(1, 8)$
- D) $(8, -1)$

Question 7

Skill: Functions — Absolute Value

The function f is defined by $f(x) = |3x - 10x|$. What value of a satisfies $f(4) - f(a) = -21$?

- A) -28
- B) 2
- C) 7
- D) 52

Question 8

Skill: Algebraic Manipulation

If $-7(12 - 4x) + 3 = -8(12 - 4x) + 15$, what is the value of $12 - 4x$?

Answer: 12

[Student-Produced Response]

Question 9

Skill: Inequalities — Word Problems

The minimum value of y is 9 less than 5 times another number p . Which inequality shows the possible values of y ?

- A) $y \leq 5p - 9$
- B) $y \geq 5p - 9$
- C) $y \leq 9 - 5p$

D) $y \geq 9 - 5p$

Question 10

Skill: Inequalities — Graphing

[Graph showing a horizontal dashed line at $y = -7$ with the region above the line shaded (i.e., $y > -7$). The x-axis spans from -8 to 8 .]

The shaded region shown represents the solutions to the inequality $-14y < c$, where c is a constant. What is the value of c ?

- A) 196
- B) 14
- C) -14
- D) 98

Question 11

Skill: Systems of Equations — Mixture

At a constant temperature, a chemist combines two solutions of different salt concentrations to create a mixture with a total volume of 200 milliliters, of which 28% is salt. Solution A contributes x milliliters to the mixture, of which 18% is salt. Solution B contributes y milliliters, of which 42% is salt. Which system of equations represents this situation?

- A) $x + y = 200$ and $0.42x + 0.18y = 0.28(200)$
- B) $x + y = 200$ and $0.18x + 0.42y = 0.28(200)$
- C) $x + y = 0.28(200)$ and $0.42x + 0.18y = 200$
- D) $x + y = 0.28(200)$ and $0.18x + 0.42y = 200$

Question 12

Skill: Exponential Functions — Graphing

[Graph showing the curve $y = 3^x + k$. The curve passes through $(0, -4)$ and has a horizontal asymptote at $y = -5$.]

The graph of the equation $y = 3^x + k$ is shown, where k is a constant. What is the value of k ?

- A) -6
- B) -5
- C) -4
- D) -3

Question 13

Skill: Ratios — Unit Conversion

A certain lake has an area of 5.12 square kilometers. What is the area, in square meters, of this lake? (1 kilometer = 1,000 meters)

- A) 512

- B) 5,120
- C) 512,000
- D) 5,120,000

Question 14

Skill: Functions — Evaluation

The function g is defined by $g(x) = 3x^2$. What is the value of $g(7)$?

- A) 21
- B) 42
- C) 63
- D) 147

Question 15

Skill: Functions — Definition

The function h gives the product of a number, x , and a number that is 48 more than x . Which equation defines h ?

- A) $h(x) = x^2 + x + 48$
- B) $h(x) = x^2 + 48$
- C) $h(x) = x^2 + 48x$
- D) $h(x) = x^2 + 48x + 48$

Question 16

Skill: Polynomials — Factoring

One of the factors of $6x^3 + 54x^2 + 120x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Answer: 4

[Student-Produced Response]

Question 17

Skill: Linear Functions — Interpretation

Jordan exercises in up to 3 ways each week for a total of 10 hours while preparing for a marathon. Jordan stretches for the same number of minutes each week. The equation $y = 600 - x - 180$ represents the situation where Jordan cycles for x minutes during a week and runs for any remaining training time y , in minutes. If this equation is graphed in the xy -plane, which statement is the best interpretation of the x -intercept?

- A) During a week when Jordan stretches for 420 minutes, she cycles for 180 minutes.
- B) Each week, Jordan cycles and runs for a total of 420 minutes.
- C) During a week when Jordan doesn't cycle, she runs for 420 minutes.
- D) During a week when Jordan doesn't run, she cycles for 420 minutes.

Question 18

Skill: Geometry — Angles

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(7x - 140)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

- A) $(-14x + 280)^\circ$
- B) $(-14x + 640)^\circ$
- C) $(14x - 280)^\circ$
- D) 180°

Question 19

Skill: Advanced Equations

$$(2/5)(5x + 15)(x + \sqrt{k + 9})(x - \sqrt{k + 9}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 45. What is the value of k ?

Answer: 6

[Student-Produced Response]

Question 20

Skill: Statistics — Standard Deviation

Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

- A) Value: 1(freq 0), 2(freq 8), 3(freq 44), 4(freq 8), 5(freq 0)
- B) Value: 1(freq 6), 2(freq 12), 3(freq 24), 4(freq 12), 5(freq 6)
- C) Value: 1(freq 10), 2(freq 10), 3(freq 20), 4(freq 10), 5(freq 10)
- D) Value: 1(freq 12), 2(freq 12), 3(freq 12), 4(freq 12), 5(freq 12)

- A) A
- B) B
- C) C
- D) D

Question 21

Skill: Geometry — Similar Solids

[Table: Right circular cone A has volume 96π cubic units. Right circular cone B has volume $2,592\pi$ cubic units.]

The table shows the volume of two similar solids, right circular cone A and right circular cone B. The radius of cone A is 3 units. The surface area of cone A is $m\pi$ square units, and the surface area of cone B is $n\pi$ square units, where m and n are constants. What is the value of $n - m$? (Surface area of a cone = $\pi r^2 + \pi rl$)

Answer: 432

[Student-Produced Response]

Question 22

Skill: Quadratic Functions

Function g is a quadratic function. The graph of $y = g(x)$ in the xy -plane has a vertex at $(-5, 0)$, contains the point $(-6, -3)$, and has a y -intercept at $(0, c)$. The graph of $y = 4$ times $g(x)$ has a y -intercept at $(0, d)$. What is the positive difference between c and d ?

Answer: 225

[Student-Produced Response]

ANSWER KEY

Section 1, Module 1 — Reading & Writing

1. C 2. D 3. D 4. B 5. D 6. B 7. B 8. D 9. C
10. A 11. D 12. D 13. D 14. D 15. D 16. A 17. C 18. A
19. C 20. B 21. A 22. D 23. B 24. A 25. B 26. A 27. D

Section 1, Module 2 — Reading & Writing

1. D 2. A 3. B 4. A 5. B 6. A 7. A 8. B 9. C
10. D 11. D 12. D 13. D 14. C 15. D 16. A 17. B 18. C
19. A 20. A 21. B 22. D 23. B 24. B 25. D 26. D 27. C

Section 2, Module 1 — Math

1. A 2. A 3. C 4. 70 5. B 6. 56 7. B 8. A 9. C
10. 11 11. D 12. B 13. C 14. B 15. A 16. 12.75 17. C 18. C
19. D 20. 0.50 21. 4 22. D

Section 2, Module 2 — Math

1. B 2. D 3. 8 4. A 5. 176 6. B 7. C 8. 12 9. B
10. D 11. B 12. B 13. D 14. D 15. C 16. 4 17. D 18. A
19. 6 20. A 21. 432 22. 225

Total Questions: 98 (54 Reading & Writing + 44 Math)

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